Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Live Oak Dairy B

Physical address of dairy:

21334 American AVE	Hilmar	Merced	95324
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1947

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X045-X170-X011-XXXX

B. OPERATORS

Nyman, Brad

Operator name: Nyman, Brad

Telephone no.: Landline: ; Cellular: (209) 495-8821

P.O. Box 1331	Hilmar	CA	65324
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Andrade, Annette

Legal owner name: Andrade, Annette

Telephone no.: Landline: (916) 000-0000; Cellular:

21334 American AVE	Hilmar	CA	95324
Mailing Address Number and Street	City	State	Zip Code

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	630	90	0	0	0	0
Number under roof	0	0	0	0	0	0
Maximum number	630	90	0	0	0	0
Average number	630	90	0	0	0	0
Avg live weight (lbs)	950	1,000	0	0		

Predominant milk cow breed: Jersey

Average milk production: 50 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 15,825.01 *tons per reporting period*

Total nitrogen from manure: 208,380.27 *lbs per reporting period*

After ammonia losses (30% loss applied): 145,866.19 *lbs per reporting period*

Total phosphorus from manure: 34,383.11 *lbs per reporting period*

Total potassium from manure: 112,372.61 *lbs per reporting period*

Total salt from manure: 317,331.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 9,540,000 *gallons*

Total nitrogen generated: 27,580.31 *lbs*

Total phosphorus generated: 4,890.61 *lbs*

Total potassium generated: 45,117.33 *lbs*

Total salt generated: 323,507.28 *lbs*

	9,540,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	9,540,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
TID Lateral 6	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/21/2023	Corral solids	1,360.00 *ton*	Dry-weight	22.1		14,500.00	2,080.00	3,840.00		48.80

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	30,723.76	4,407.27	8,136.50	1,034,013.44
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	30,723.76	4,407.27	8,136.50	1,034,013.44

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	8	8	2	process wastewater	X045-X170-X048-XXXX X045-X170-X049-XXXX
2	12	12	2	process wastewater	X045-X170-X011-XXXX X045-X170-X012-XXXX
3	38	38	2	process wastewater	X045-X170-X012-XXXX
Totals for areas that were used for application	58	58	6		
Totals for areas that were not used for application					
Land application area totals	58	58	6		

B. CROPS AND HARVESTS

1

Field name: 1

11/12/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 8 Plant date: 11/12/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/11/2023	166.00 *ton*	As-is		70.0	6,240.00	1,470.00	15,900.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	20.75	258.96	61.01	659.85	1,245.00

1

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 8 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	220.00 *ton*	As-is		69.4	3,490.00	860.00	5,620.00		4.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	27.50	191.95	47.30	309.10	706.86

2

Field name: 2

11/12/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 12 Plant date: 11/12/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/11/2023	244.00 *ton*	As-is		70.0	5,640.00	1,500.00	1,200.00		10.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	20.33	229.36	61.00	48.80	1,256.60

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 12 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	328.00 *ton*	As-is		65.2	3,450.00	630.00	4,960.00		1.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	27.33	188.60	34.44	271.15	323.41

3

Field name: 3

11/12/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 38 Plant date: 11/12/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/11/2023	770.00 *ton*	As-is		70.0	6,510.00	1,650.00	16,800.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	20.26	263.83	66.87	680.84	1,240.11

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	1,110.00 *ton*	As-is		68.8	3,150.00	610.00	4,760.00		3.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	29.21	184.03	35.64	278.08	601.50

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 11/12/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough Plant date: 11/12/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.07	10.14	82.62	961.34	180,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	217.99	972,000.00 *gal*
Application event totals		51.07	10.14	82.62	1,179.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	97.32	22.94	143.33	1,220.46	300,000.00 *gal*
Application event totals		97.32	22.94	143.33	1,220.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.13	16.06	100.33	854.32	210,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	254.32	1,134,000.00 *gal*
Application event totals		68.13	16.06	100.33	1,108.64	

1 - 06/01/2023: Corn, silage

Field name: 1

Crop: Corn, silage Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

1 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	98.76	8.41	191.52	1,089.02	180,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	217.99	972,000.00 *gal*
Application event totals		98.76	8.41	191.52	1,307.01	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	290.66	1,296,000.00 *gal*
Application event totals		0.00	0.00	0.00	290.66	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	98.76	8.41	191.52	1,089.02	180,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	217.99	972,000.00 *gal*
Application event totals		98.76	8.41	191.52	1,307.01	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	145.33	648,000.00 *gal*
Application event totals		0.00	0.00	0.00	145.33	

07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.99	10.45	41.46	219.06	150,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	181.66	810,000.00 *gal*
Application event totals		35.99	10.45	41.46	400.72	

1 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	181.66	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	181.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	217.99	972,000.00 *gal*
Application event totals		0.00	0.00	0.00	217.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	28.79	8.36	33.17	175.25	120,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	145.33	648,000.00 *gal*
Application event totals		28.79	8.36	33.17	320.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	217.99	972,000.00 *gal*
Application event totals		0.00	0.00	0.00	217.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	217.99	972,000.00 *gal*
Application event totals		0.00	0.00	0.00	217.99	

2 - 11/12/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough

Plant date: 11/12/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

2 - 11/12/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	56.75	11.27	91.80	1,068.16	300,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	242.21	1,620,000.00 *gal*
Application event totals		56.75	11.27	91.80	1,310.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	103.81	24.47	152.88	1,301.82	480,000.00 *gal*
Application event totals		103.81	24.47	152.88	1,301.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.88	15.29	95.55	813.64	300,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	242.21	1,620,000.00 *gal*
Application event totals		64.88	15.29	95.55	1,055.85	

2 - 06/01/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	87.79	7.48	170.24	968.02	240,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	193.77	1,296,000.00 *gal*
Application event totals		87.79	7.48	170.24	1,161.79	

2 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	290.66	1,944,000.00 *gal*
Application event totals		0.00	0.00	0.00	290.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	109.74	9.35	212.80	1,210.03	300,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	242.21	1,620,000.00 *gal*
Application event totals		109.74	9.35	212.80	1,452.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	193.77	1,296,000.00 *gal*
Application event totals		0.00	0.00	0.00	193.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.39	11.15	44.23	233.66	240,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	193.77	1,296,000.00 *gal*
Application event totals		38.39	11.15	44.23	427.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	242.21	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	242.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	169.55	1,134,000.00 *gal*
Application event totals		0.00	0.00	0.00	169.55	

2 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	28.79	8.36	33.17	175.25	180,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	145.33	972,000.00 *gal*
Application event totals		28.79	8.36	33.17	320.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	121.11	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	121.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	145.33	972,000.00 *gal*
Application event totals		0.00	0.00	0.00	145.33	

3 - 11/12/2022: Oats, silage-soft dough

Field name: 3

Crop: Oats, silage-soft dough

Plant date: 11/12/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.76	10.67	86.96	1,011.94	900,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	229.47	4,860,000.00 *gal*
Application event totals		53.76	10.67	86.96	1,241.41	

3 - 11/12/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	98.35	23.18	144.83	1,233.30	1,440,000.00 *gal*
Application event totals		98.35	23.18	144.83	1,233.30	

02/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.37	13.52	84.49	719.43	840,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	214.17	4,536,000.00 *gal*
Application event totals		57.37	13.52	84.49	933.59	

3 - 06/01/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	97.03	8.26	188.16	1,069.92	840,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	214.17	4,536,000.00 *gal*
Application event totals		97.03	8.26	188.16	1,284.08	

06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	290.66	6,156,000.00 *gal*
Application event totals		0.00	0.00	0.00	290.66	

3 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	97.03	8.26	188.16	1,069.92	840,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	214.17	4,536,000.00 *gal*
Application event totals		97.03	8.26	188.16	1,284.08	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	275.36	5,832,000.00 *gal*
Application event totals		0.00	0.00	0.00	275.36	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	36.37	10.56	41.90	221.36	720,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	183.57	3,888,000.00 *gal*
Application event totals		36.37	10.56	41.90	404.93	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	260.06	5,508,000.00 *gal*
Application event totals		0.00	0.00	0.00	260.06	

08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	183.57	3,888,000.00 *gal*
Application event totals		0.00	0.00	0.00	183.57	

08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	229.47	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	229.47	

3 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.31	8.80	34.92	184.47	600,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	152.98	3,240,000.00 *gal*
Application event totals		30.31	8.80	34.92	337.45	

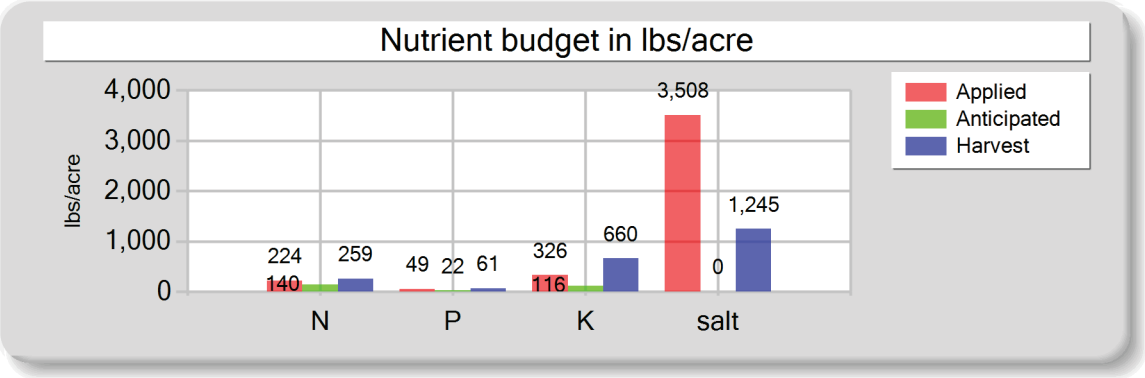
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	183.57	3,888,000.00 *gal*
Application event totals		0.00	0.00	0.00	183.57	

B. NUTRIENT BUDGET

1 - 11/12/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 11/12/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	216.52	49.13	326.27	3,036.12
Fresh water	0.00	0.00	0.00	472.32
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	223.52	49.13	326.27	3,508.44
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	258.96	61.01	659.85	1,245.00
Nutrient balance	-35.44	-11.87	-333.58	2,263.44
Applied to removed ratio	0.86	0.81	0.49	2.82

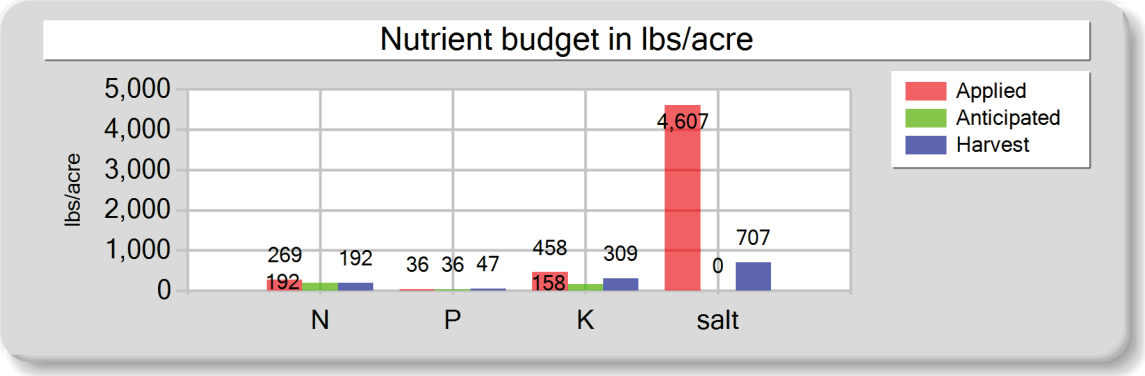
Fresh water applied	
2,106,000.00	*gallons*
77.56	*acre-inches*
9.69	*inches/acre*

Process wastewater applied	
690,000.00	*gallons*
25.41	*acre-inches*
3.18	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 - 06/01/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	262.30	35.64	457.67	2,572.35
Fresh water	0.00	0.00	0.00	2,034.59
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	269.30	35.64	457.67	4,606.94
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	191.95	47.30	309.10	706.86
Nutrient balance	77.35	-11.66	148.57	3,900.08
Applied to removed ratio	1.40	0.75	1.48	6.52

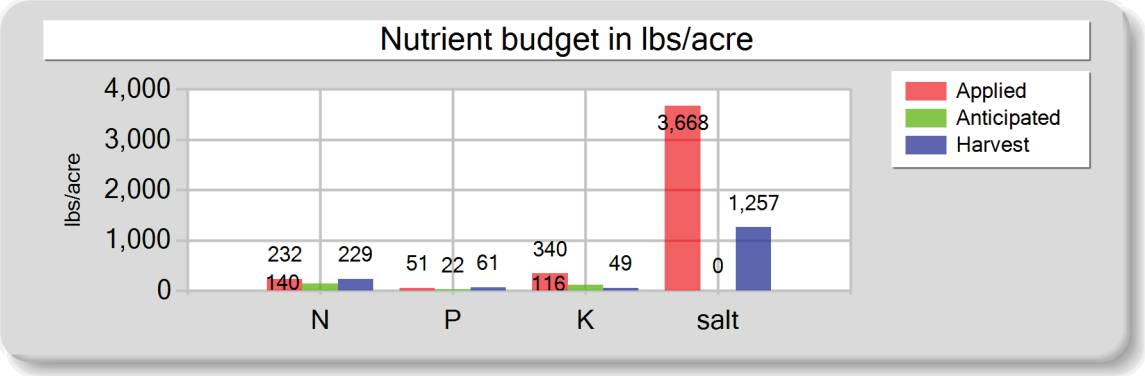
Fresh water applied	
9,072,000.00	*gallons*
334.09	*acre-inches*
41.76	*inches/acre*

Process wastewater applied	
630,000.00	*gallons*
23.20	*acre-inches*
2.90	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 11/12/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 11/12/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	225.44	51.03	340.23	3,183.62
Fresh water	0.00	0.00	0.00	484.43
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	232.44	51.03	340.23	3,668.04
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	229.36	61.00	48.80	1,256.60
Nutrient balance	3.08	-9.97	291.43	2,411.44
Applied to removed ratio	1.01	0.84	6.97	2.92

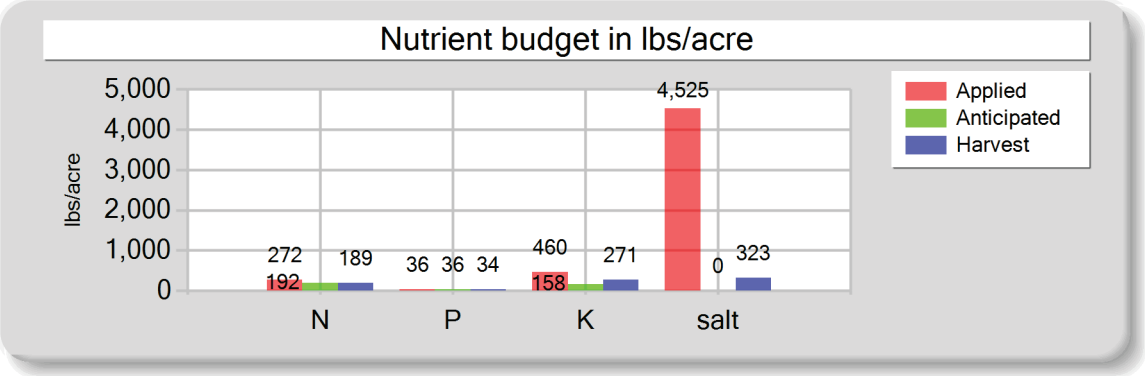
Fresh water applied	
3,240,000.00	*gallons*
119.32	*acre-inches*
9.94	*inches/acre*

Process wastewater applied	
1,080,000.00	*gallons*
39.77	*acre-inches*
3.31	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 06/01/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	264.70	36.33	460.44	2,586.95
Fresh water	0.00	0.00	0.00	1,937.71
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	271.70	36.33	460.44	4,524.66
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	188.60	34.44	271.15	323.41
Nutrient balance	83.10	1.89	189.29	4,201.25
Applied to removed ratio	1.44	1.05	1.70	13.99

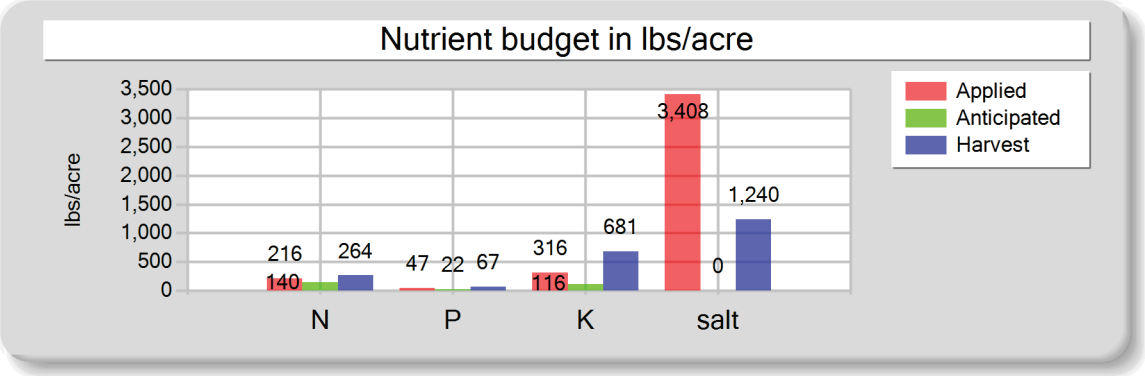
Fresh water applied	
12,960,000.00	*gallons*
477.27	*acre-inches*
39.77	*inches/acre*

Process wastewater applied	
960,000.00	*gallons*
35.35	*acre-inches*
2.95	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 11/12/2022: Oats, silage-soft dough

Field name: 3 Crop: Oats, silage-soft dough Plant date: 11/12/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	209.48	47.37	316.28	2,964.67
Fresh water	0.00	0.00	0.00	443.63
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	216.48	47.37	316.28	3,408.30
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	263.83	66.87	680.84	1,240.11
Nutrient balance	-47.35	-19.49	-364.56	2,168.20
Applied to removed ratio	0.82	0.71	0.46	2.75

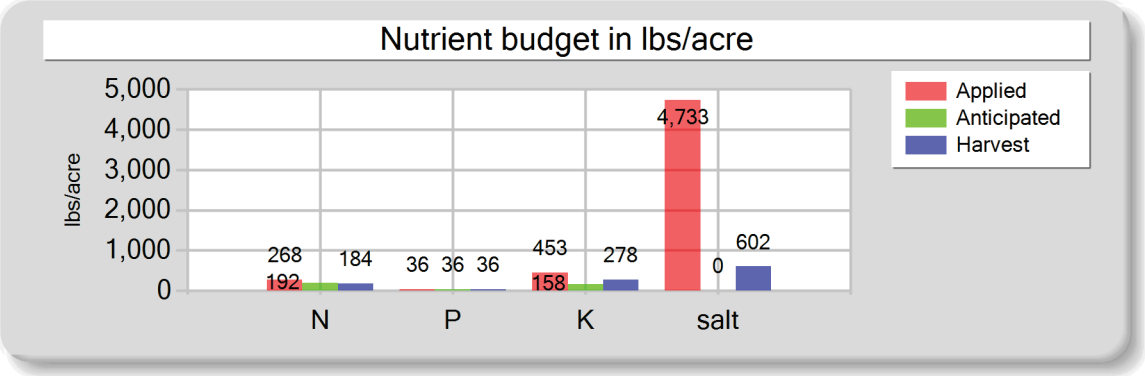
Fresh water applied	
9,396,000.00	*gallons*
346.02	*acre-inches*
9.11	*inches/acre*

Process wastewater applied	
3,180,000.00	*gallons*
117.11	*acre-inches*
3.08	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 06/01/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	260.73	35.89	453.13	2,545.66
Fresh water	0.00	0.00	0.00	2,187.57
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	267.73	35.89	453.13	4,733.24
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	184.03	35.64	278.08	601.50
Nutrient balance	83.71	0.26	175.05	4,131.73
Applied to removed ratio	1.45	1.01	1.63	7.87

Fresh water applied	
46,332,000.00	*gallons*
1,706.25	*acre-inches*
44.90	*inches/acre*

Process wastewater applied	
3,000,000.00	*gallons*
110.48	*acre-inches*
2.91	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 05/01/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 22.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,500.00	2,080.00	3,840.00							48.80
DL	100.00	100.00	100.00							0.01

Separaated Solids

Sample and source description: Separaated Solids

Sample date: 10/24/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,600.00	2,000.00	3,320.00	6,130.00	2,090.00	1,060.00	2,070.00	411.00		17.60
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

B. PROCESS WASTEWATER ANALYSES

WWQ4

Sample and source description: WWQ4

Sample date: 12/12/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	272.00	174.00	0.00	0.00	54.00	440.00								4,150.00	5,120
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ1

Sample and source description: WWQ1

Sample date: 03/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	311.00	204.00	0.00	0.00	73.30	458.00								4,360.00	3,900
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 06/06/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	526.00	378.00	0.00	0.00	44.80	1,020.00								7,640.00	5,800
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 09/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	230.00	120.00	0.00	0.00	66.80	265.00	78.90	57.20	64.10	1,120.00	0.00	81.40	102.00	2,670.00	1,400
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

C. FRESH WATER ANALYSES

TID Lateral 6

TID Lateral 6

TID lateral 6

Sample description: TID lateral 6

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	215
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

1

Soil

Sample and source description: Soil

Sample date: 11/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			38.00				
DL			4.00				

2

Soil

Sample and source description: Soil

Sample date: 11/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			45.00				
DL			4.00				

3

3

Soil

Sample and source description: Soil

Sample date: 11/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			51.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

1 - 11/12/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,240.00	1,470.00	15,900.00		10.00
DL	100.00	100.00	100.00		0.01

1 - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,490.00	860.00	5,620.00		4.20
DL	100.00	100.00	100.00		0.01

2 - 11/12/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,640.00	1,500.00	1,200.00		10.30
DL	100.00	100.00	100.00		0.01

2 - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,450.00	630.00	4,960.00		1.70
DL	100.00	100.00	100.00		0.01

3 - 11/12/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,510.00	1,650.00	16,800.00		10.20
DL	100.00	100.00	100.00		0.01

3 - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,150.00	610.00	4,760.00		3.30
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

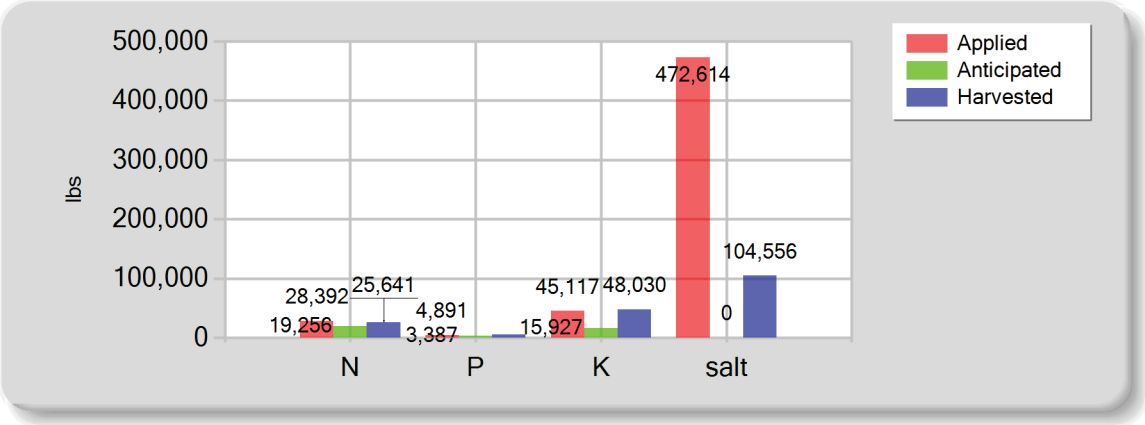
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

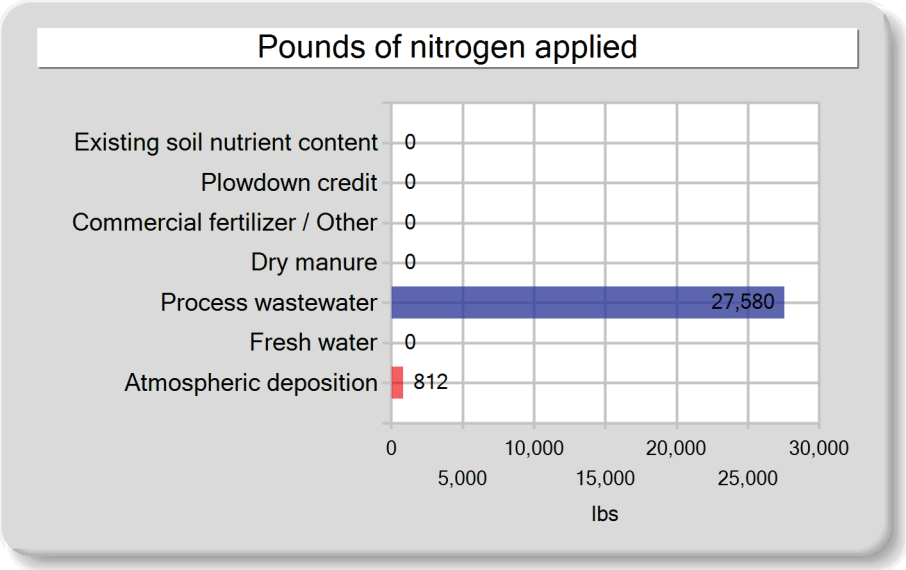
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

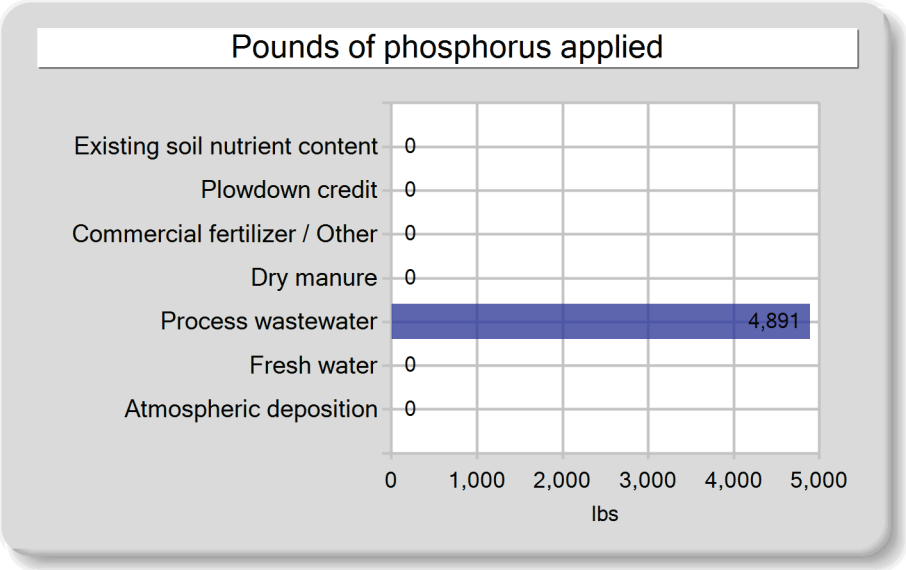
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	27,580.31	4,890.61	45,117.33	323,507.28
Fresh water	0.00	0.00	0.00	149,106.71
Atmospheric deposition	812.00	0.00	0.00	0.00
Total nutrients applied	28,392.31	4,890.61	45,117.33	472,613.98
Anticipated crop nutrient removal	19,256.00	3,387.20	15,926.80	0.00
Actual crop nutrient removal	25,641.20	5,906.92	48,030.16	104,556.10
Nutrient balance	2,751.11	-1,016.31	-2,912.83	368,057.89
Applied to removed ratio	1.11	0.83	0.94	4.52

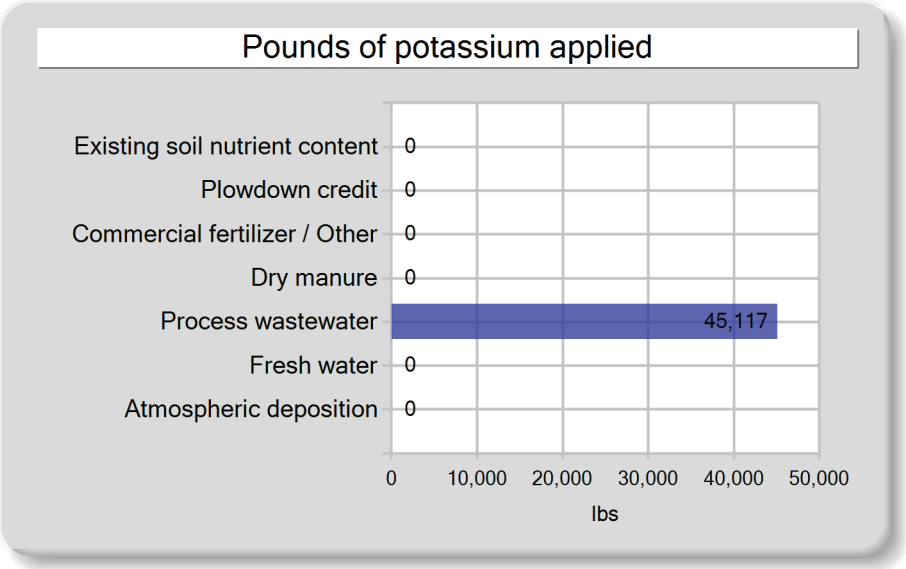
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

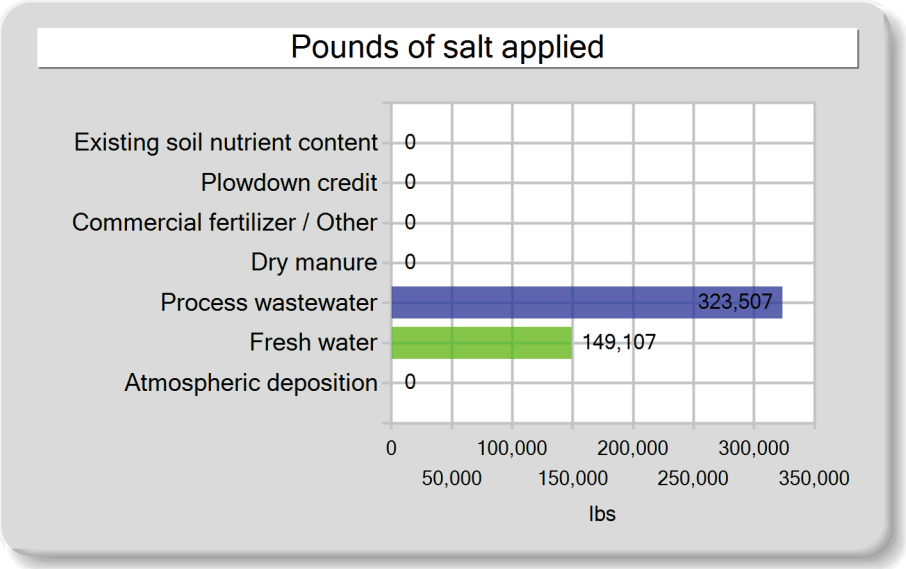


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Annette Andrade	Brad Nyman
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Annette Andrade	Brad Nyman
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6/19/24	6-19-24
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted.

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed.

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Brad Nyman

Name of Dairy Facility: Live Oak Dairy B

Facility Address:

Number and Street	City	County	Zip Code
21334 American AVE	Hilmar	Merced	95324

Contact Person Name and Phone Number:

Name	Phone Number
Brad Nyman	(209) 000-0000

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Silva & Sons Custom Spreading

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
P.O. Box 1261	Hilmar	CA	95324

Contact Person:

Name	Phone Number
Frank Silva	(209) 632-8202

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Broker

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Phone Number
Frank Silva	(209) 632-8202

Address	City	State	Zip Code
23616 Williams AVE	Hilmar	CA	95324

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code
23616 Williams AVE	Hilmar	95324

Street and nearest cross street (if no address)	County
	Merced

Assessor's Parcel Number	Assessor's Parcel Number County

Last date hauled: 05/21/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 1,360.00 tons

Manure Solids Content: 77.9 %

Method used to determine amount of manure:

Haulers Calcs.

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Brad Nyman
Brad Nyman (Jun 19, 2024 09:29 PDT)
Operator Signature

Jun 19, 2024
Date

Frank Silva
Hauler Signature

Date

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Live Oak Dairy B
P.O. Box 1331
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/28/23 1:36 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/28/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 14.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW	T3240026A	Electrical Conductivity (EC)	EPA 120.1 and	1.45	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	43.5	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	930	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	97.8	95.2 (T3240023E)	96.4 (T3240023E)	1.25
Electrical Conductivity (EC)	100			0.217
Nitrate Nitrogen (NO3-N)	94.9	105 (T3240022E)	103 (T3240022E)	2.54

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Live oak B
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: FR
Sampled By: BN
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	8/25	12:45	Domestic well	DW	14.4°		✓				
2											
3											
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
1st		FRAG				
2nd		DENELE ANALYTICAL, INC.	8-28-23 12:36			T3240026

Required to be submitted to any agency? (YES / NO)